A very long title for my slides, which may span across two lines

Gianluca Baio

g.baio@ucl.ac.uk
https://gianluca.statistica.it | https://egon.stats.ucl.ac.uk/research/statistics-health-economics
https://github.com/giabaio | https://github.com/StatisticsHealthEconomics
@gianlubaio@mas.to | gianluca-baio

Follow our departmental social media accounts + magazine "Sample Space"

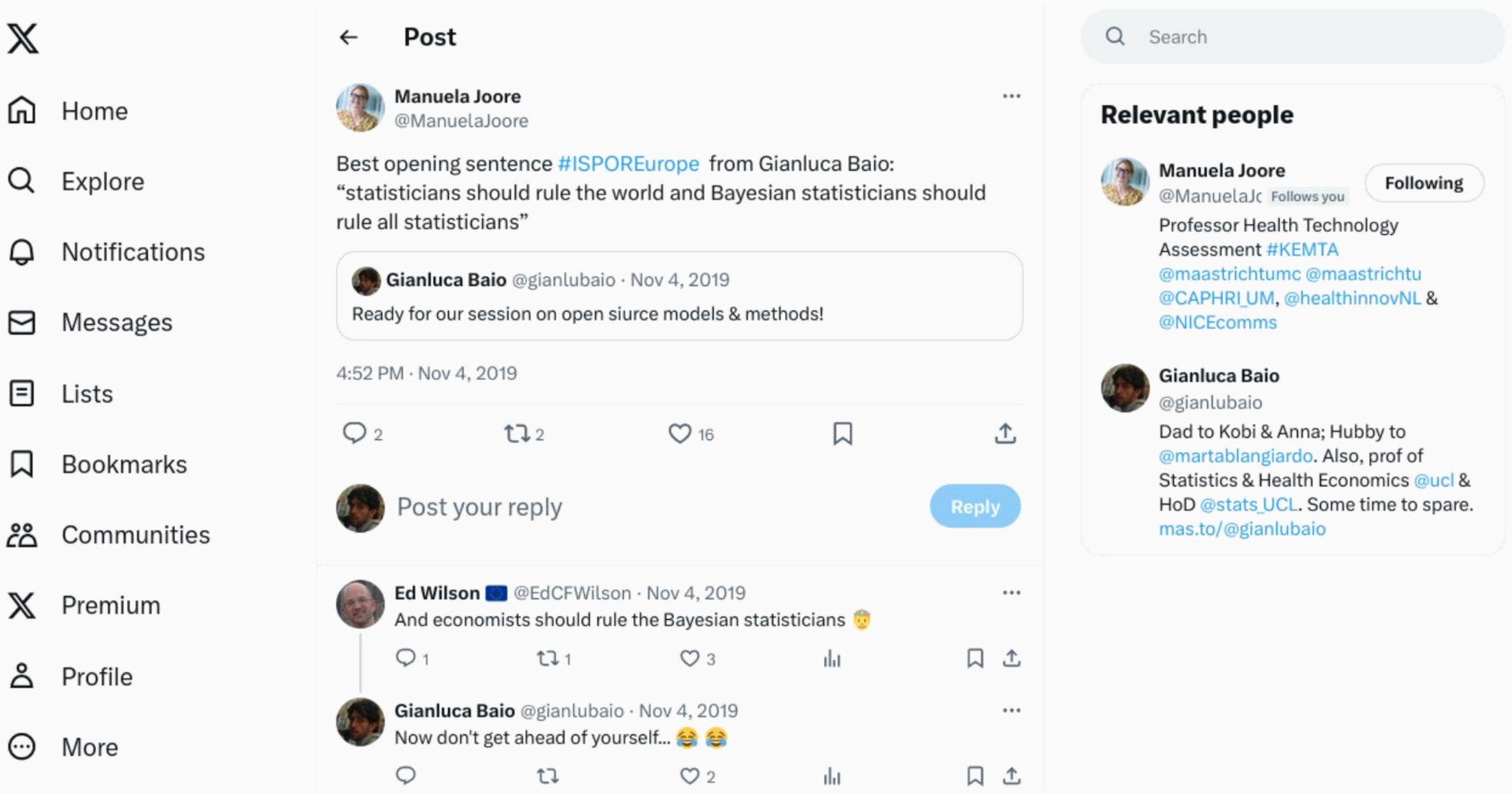
13 October 2026

The name of the session
The name of the conference, Where it is

check out our departmental podcast "Random Talks" on Soundcloud!

Department of Statistical Science | University College London

Disclaimer...



... Just so you know what you're about to get yourself into... 😉

Hello there 👋

This presentation will show you examples of what you can do with Quarto and Reveal.js, including:

- Presenting code and LaTeX equations
- Including computations in slide output
- Image, video, and iframe backgrounds
- Fancy transitions and animations
- Printing to PDF
  - Which is sometimes useful...

...and much more

Different output

- The default output is `html` via `revealjs`. This is managed by the option

```
format:
   ucl-revealjs
```

  in the `yml` of the `.qmd` file.

- Alternatively, the slides can be produced in `pdf` directly via `beamer`. This is obtained by specifying

```
format:
   ucl-beamer
```

  instead.

Printing to pdf via `decktape`

This is probably the best option. Typeset and render the slides to `html`, to take full advantage of the `quarto`/`revealjs` features, then use `decktape` to print the `html` file to `pdf`

- `decktape` maintains the main features of the `html` and the result is very good

Here's a new slide without a title.

No logo

Here's a slide with no logo – just add `{.no-logo}` to the title of the slide

Embedding sites

- Can also embed websites as `iframes`, using the code

```
## {background-iframe="https://gianluca.statistica.it" background-interactive="true" data-preload="true" .no-logo}
```

- This creates a title-less slide with the full space occupied by the website.
  - Use the options `background-interactive="true"` and `data-preload="true"` for best control and maximum load speed

Fun with colours & spaces

With a subtitle

This shows how to use colours in the slides. A bunch are pre-defined in the `.css` file and can be used simply as:

- The **predictive** distribution can be computed as follows.

More can be created by simply modifying the `.css` file to add more `HEX` codes.

- For example use

```
.uclblue {
  color: #002855;
}
```

(though this colour is already defined)

Can add vertical space using `css` styles

```
::: {style="margin-top: 40px"}
Can add vertical space using `css` styles
:::
```

Some maths

We can also include maths, using standard LaTeX:

$$p(y \mid \mathcal{D}) = \int p(y \mid \boldsymbol{\theta}) p(\boldsymbol{\theta} \mid \mathcal{D}) d\boldsymbol{\theta}$$

To use colours with LaTeX, needs a little hack: `$\class{color}{expression}$`

You can define a probabilistic assumption as

$$y \mid \boldsymbol{\theta} \sim \text{Poisson}(\boldsymbol{\theta})$$

but in line it would look like this $y \sim \text{Normal}(\mu, \sigma^2)$

These use the predefined LaTeX macros, saved in the file `latex_macros.html`. More can be (re)defined by simply adding to the file, which gets pre-loaded at the beginning of the resulting `.html` output; example: `$\DIC=2345$` is typeset as DIC $= 2345$.

For instance, the command `$\dnorm$` translates to the text Normal (typeset in the current font). But can also use `$\txt{Some text}$`, which gets rendered as Some text– it's like `\text{...}` but with the current font

More maths

The font used to typeset numbers in maths mode can be controlled by the `yml` option `math-numbers-font` (default to `false`).

- When this is set to `true`, then the numbers in maths mode inherit the main font (UCL Sans)
- This will look like

$$\int_0^\infty x^4 p(x \mid y_1) dx \qquad \text{and} \qquad \text{DIC} = 1234.$$

  - NB: the font is controlled by the definition of the macro `\mnum{}` in the `./assets/latex_macros.html` file

Pretty Code

(Slide 1)

- Over 20 syntax highlighting themes available
- Default theme optimized for accessibility

```r
# Define a server for the Shiny app
function(input, output) {

  # Fill in the spot we created for a plot
  output$phonePlot <- renderPlot({
    # Render a barplot
  })
}
```

Controlling the font-size of code chunks

This can be done by modifying the value set in the `./css/common.css` file:

```scss
.reveal div.sourceCode {
  margin: 0;
  overflow: auto;
  border-color: $link-color;
  font-size: 1.40em;
}

/* This controls the font size of the output of R chunks */
.reveal pre:not(.sourceCode) {
  font-size: .75em;
}
```

Code Animations

(Slide 2)

- Over 20 syntax highlighting themes available
- Default theme optimized for accessibility

```
# Define a server for the Shiny app
function(input, output) {

  # Fill in the spot we created for a plot
  output$phonePlot <- renderPlot({
    # Render a barplot
    barplot(WorldPhones[,input$region]*1000,
            main=input$region,
            ylab="Number of Telephones",
            xlab="Year")
  })
}
```

More on animation, next

Auto-animation

$$y \sim \text{Normal}(\mu, \sigma)$$

Auto-animation

$$y \sim \text{Normal}(\mu, \sigma)$$

$$\mu = \beta_0 + \beta_1 x$$

Auto-animation

When two consecutive slides both have the `{auto-animate=true}` in the title, `reveal.js` looks for elements that appear on both slides and animates each one from its position, size and style on the first slide to those on the second, instead of switching abruptly.

Going from the first slide to the second, the equation slides up into its new place and the new line fades in, so it looks like one slide growing rather than two separate slides.

Elements found on only one of the two slides fade in or out. Some useful extra options on the slide:

- `{auto-animate-duration=0.5}` sets the duration in seconds.
- `{auto-animate-easing="ease-in-out"}` sets the speed curve.
- `{auto-animate-restart=true}` breaks a chain, so that slide doesn't animate from the previous one.

(More) Auto-animation

Auto-animate tries to automatically find the bits to match across slides. To override, see: https://quarto.org/docs/presentations/revealjs/advanced.html#element-matching

Automatically animate matching elements across slides with Auto-Animate.



(More) Auto-animation

Automatically animate matching elements across slides with Auto-Animate.



Learn more: Auto-Animate

Line Highlighting

- Highlight specific lines for emphasis
- Incrementally highlight additional lines
- This is controlled by the chunk option `code-line-numbers=""`

```
1  import numpy as np
2  import matplotlib.pyplot as plt
3
4  r = np.arange(0, 2, 0.01)
5  theta = 2 * np.pi * r
6  fig, ax = plt.subplots(subplot_kw={'projection': 'polar'})
7  ax.plot(theta, r)
8  ax.set_rticks([0.5, 1, 1.5, 2])
9  ax.grid(True)
10 plt.show()
```

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10 plt.show()
```

Text formatting

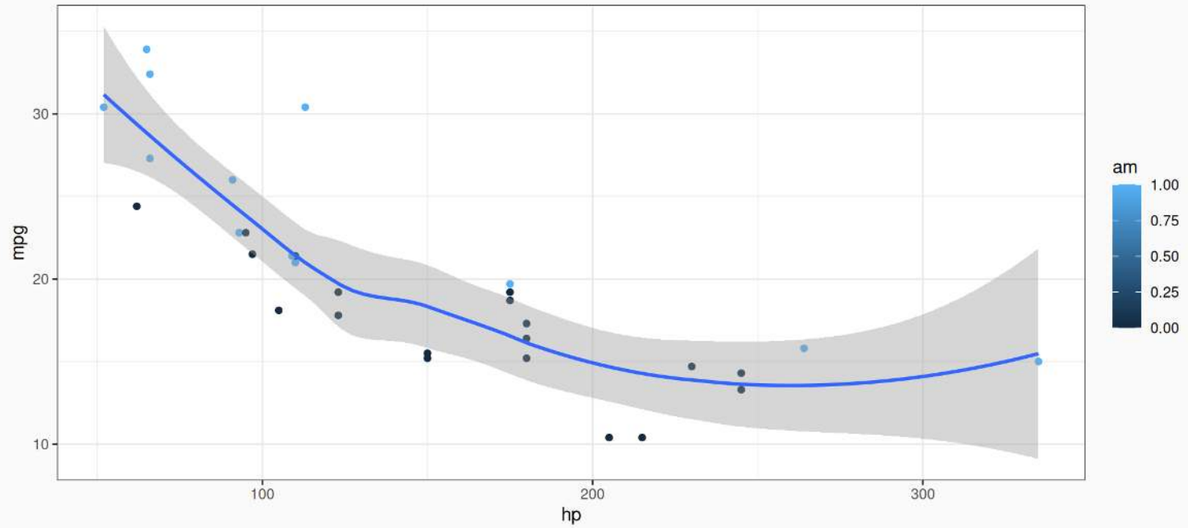
> Blockquote
>
> More quoted text

An aside at the bottom of the page, can be styled with

```
::: {.aside}
CONTENT
:::
```

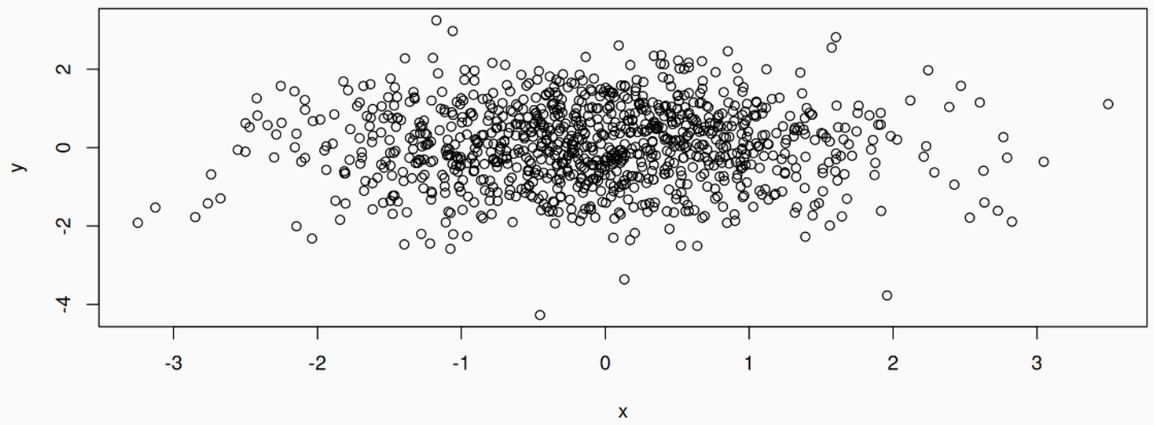
Executable Code

```
1 library(ggplot2)
2 ggplot(mtcars, aes(hp, mpg, color = am)) +
3   geom_point() +  geom_smooth(formula = y ~ x, method = "loess")
```



A base plot

```
1 plot(rnorm(1000),rnorm(1000),xlab="x",ylab="y")
```



Temporarily changing the size of code output

```r
rnorm(1000) |> summary()
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
-3.0728 -0.5984  0.0299  0.0601  0.7523  2.9870 
```

This is achieved adding the options

```
#| attr-source: 'style="font-size: 2.0em;"'
#| attr-output: 'style="font-size: 1.25em;"'
```

to that specific `R` chunk setup

A section

Using the tag `{.divider}` makes for some nicer `css` formatting (especially under `ucl-revealjs`)

And another

Using the tag `{.divider-bg}` gives a background in line with the UCL 2026 template

The slide looks nicer without the footer. This can be done either by adding `::: {.footer style="background-color: transparent!important; background: transparent!important"}` `:::` to the end of the slide, to ensure that the footer doesn't interfere with the colours...

More simply, can explicitly use the tag `{.divider-bg footer="false"}`

Section title slide

Setting the background colour to a dark colour automatically changes the colour of the text to make it visible over a dark background!

- The footer may or may not be included and its positioning/formatting can be customised

Code chunk and output side by side

```
ggplot(palmerpenguins::penguins,
       aes(x = bill_length_mm,
           y = bill_depth_mm,
           color = species)) +
  geom_point() +
  colorblindr::scale_color_OkabeIto()
```

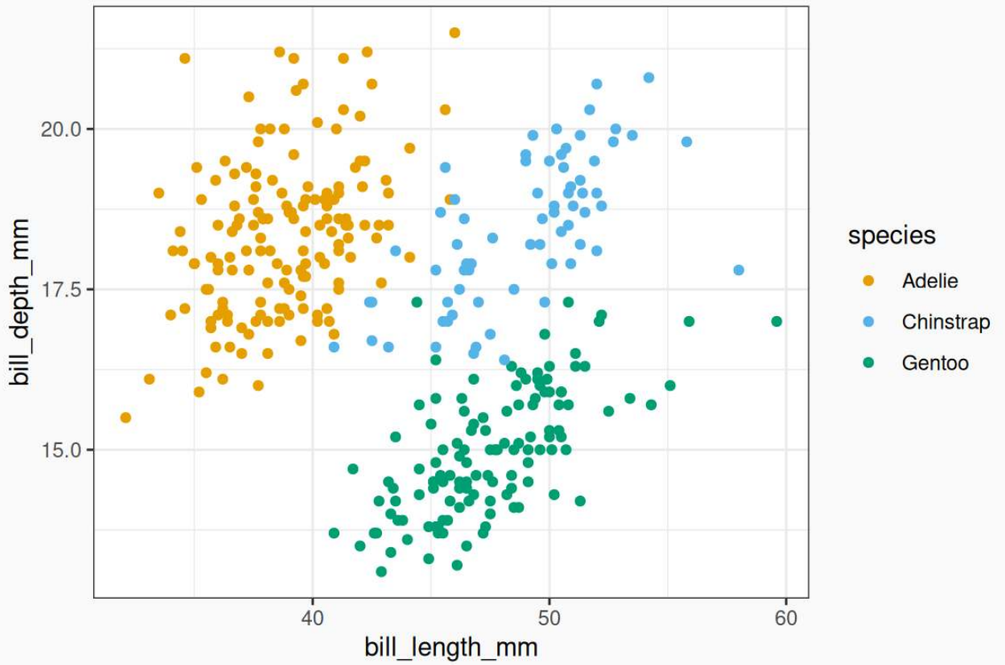
Note

This is controlled by the chunk option `output-location:`, which can take values `column-fragment` (code first, then output becomes visible), `column` (both visible at the same time), `fragment` (output appears on the same slide, but hidden until the next fragment), or `slide` (output pushed to a brand new slide)

Also, by default, `ggplot` uses a specialised `theme_ucl()` function (defined in `.assets/setup.R`) to format the graphs. This aligns with the background colour of the slides.

Code chunk and output side by side

```
ggplot(palmerpenguins::penguins,
       aes(x = bill_length_mm,
           y = bill_depth_mm,
           color = species)) +
  geom_point() +
  colorblindr::scale_color_OkabeIto()
```



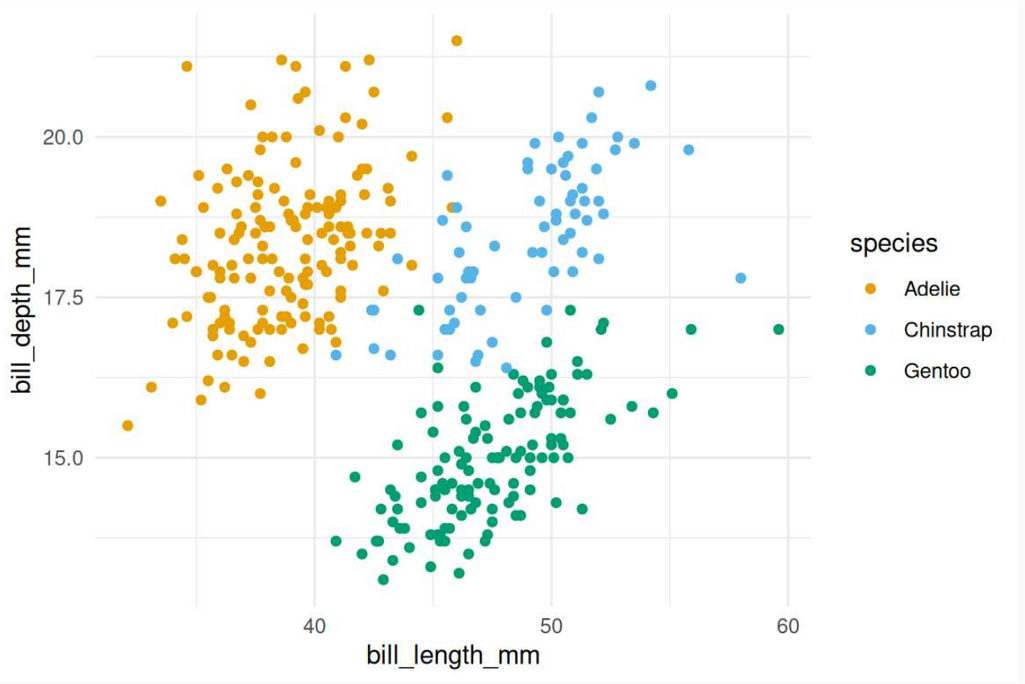
Note

This is controlled by the chunk option `output-location:`, which can take values `column-fragment` (code first, then output becomes visible), `column` (both visible at the same time), `fragment` (output appears on the same slide, but hidden until the next fragment), or `slide` (output pushed to a brand new slide)

Also, by default, `ggplot` uses a specialised `theme_ucl()` function (defined in `.assets/setup.R`) to format the graphs. This aligns with the background colour of the slides.

Code chunk and output side by side

```
ggplot(palmerpenguins::penguins,
       aes(x = bill_length_mm,
           y = bill_depth_mm,
           color = species)) +
  geom_point() +
  theme_minimal() +
  colorblindr::scale_color_OkabeIto()
```



LaTeX Equations

See code for creating columns

MathJax rendering of equations to HTML

```latex
\begin{gather*}
a_1=b_1+c_1\\
a_2=b_2+c_2-d_2+e_2
\end{gather*}

\begin{align}
a_{11}& =b_{11}&
  a_{12}& =b_{12}\\
a_{21}& =b_{21}&
  a_{22}& =b_{22}+c_{22}
\end{align}
```

$$a_1 = b_1 + c_1$$
$$a_2 = b_2 + c_2 - d_2 + e_2$$

$$a_{11} = b_{11} \qquad a_{12} = b_{12}$$
$$a_{21} = b_{21} \qquad a_{22} = b_{22} + c_{22}$$

Make all the text smaller

All the text on this slide is smaller. Not entirely sure what the use case is! Maybe long quotes?

“Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.”

Column Layout

Arrange content into columns of varying widths:

Motor Trend Car Road Tests

The data were extracted from the 1974 Motor Trend US magazine and comprise fuel consumption and 10 aspects of automobile design and performance for 32 automobiles.

mpg	cyl	disp	hp	wt
21.0	6	160	110	2.620
21.0	6	160	110	2.875
22.8	4	108	93	2.320
21.4	6	258	110	3.215
18.7	8	360	175	3.440
18.1	6	225	105	3.460

NB: Can use the package `tinytable` to format tables in a very efficient way...

Incremental Lists

Lists can optionally be displayed incrementally:

Incremental Lists

Lists can optionally be displayed incrementally:

1. First item

Incremental Lists

Lists can optionally be displayed incrementally:

1. First item
2. Second item

Incremental Lists

Lists can optionally be displayed incrementally:

1. First item
2. Second item
3. Third item

Incremental Lists

Lists can optionally be displayed incrementally:

1. First item
2. Second item
3. Third item

Insert pauses to make other types of content display incrementally.

Fragments

Incremental text display and animation with fragments:

Extra fragment styling / nesting / ordering options

Highlight custom colour (defined in scss file or can be added on the fly)

````
```{css}
#| echo: false
.reveal .slides section .fragment.highlight-itared {
  opacity: 1;
  visibility: inherit;
}
.reveal .slides section .fragment.highlight-itared.visible {
  color: #ce2b37;
}
```

::: {.fragment .highlight-itared}
Text to highlight here...
:::
````

Fragments

Incremental text display and animation with fragments:

Extra fragment styling / nesting / ordering options

Fade in

Highlight custom colour (defined in scss file or can be added on the fly)

````
```{css}
#| echo: false
.reveal .slides section .fragment.highlight-itared {
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Fragments

Incremental text display and animation with fragments:

Extra fragment styling / nesting / ordering options

Fade in

Slide up while fading in

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Incremental text display and animation with fragments:

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Slide left while fading in

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````

Fragments

Incremental text display and animation with fragments:

Extra fragment styling / nesting / ordering options

Fade in

Slide up while fading in

Slide left while fading in

Fade in then semi out

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Fragments

Incremental text display and animation with fragments:

Extra fragment styling / nesting / ordering options

Fade in

Slide up while fading in

Slide left while fading in

Fade in then semi out

Strike

Highlight red (but green and blue are also built-in)

Highlight custom colour (defined in scss file or can be added on the fly)

````
```{css}
#| echo: false
.reveal .slides section .fragment.highlight-itared {
  opacity: 1;
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.reveal .slides section .fragment.highlight-itared.visible {
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```

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Text to highlight here...
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Text to highlight here...
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  color: #ce2b37;
}
```

::: {.fragment .highlight-itared}
Text to highlight here...
:::
````

Frame-Controlled Fragments

- Some initial bullet point
- And another
  - Possibly nested...

Frame-Controlled Fragments

- Some initial bullet point
- And another
  - Possibly nested...
- Some text here
  - And some other

Frame-Controlled Fragments

- Some initial bullet point
- And another
  - Possibly nested...
- Some text here
  - And some other
- Some more text

Frame-Controlled Fragments

- Some initial bullet point
- And another
  - Possibly nested...
- More still
- And that's it

Frame-Controlled Fragments

- Some initial bullet point
- And another
  - Possibly nested...
- More still
- And that's it

Controlling fragments

- `::: {.fragment .fade-out fragment-index=x ...}` shows the content up to fragment `x`, when it is replaced by other content and just fades out
- `::: {.fragment .fade-in fragment-index=x ...}` introduces the content in fragment `x` and leaves it there
- `::: {.fragment .current-visible fragment-index=x ...}` makes the content visible *only* in fragment `x`
- Fragments can be nested, so

```
:::: {.fragment .fade-out fragment-index=3}
::: {.fragment .fade-in fragment-index=1}
Content here
:::
::::
```

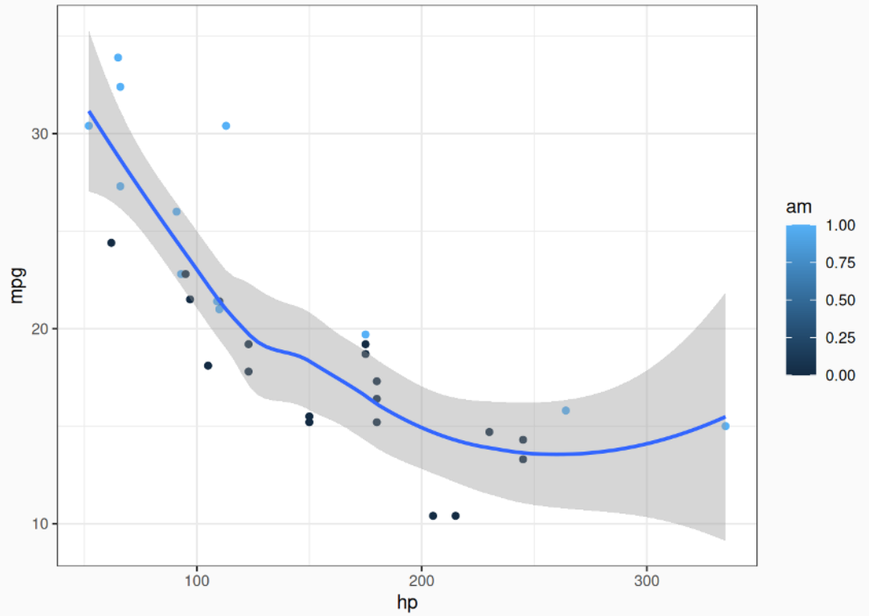
will make `Content here` appear in fragment 1 and disappear in fragment 3

Fragments and stacks

`r-stack` can be used together with fragments to center and place multiple elements on top of each other and reveal them incrementally.

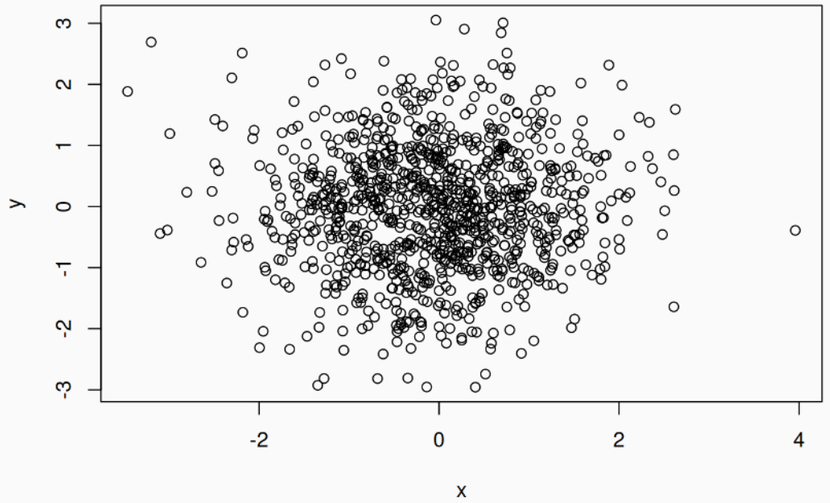
Fragments and stacks

`r-stack` can be used together with fragments to center and place multiple elements on top of each other and reveal them incrementally.



Fragments and stacks

`r-stack` can be used together with fragments to center and place multiple elements on top of each other and reveal them incrementally.



- Can style the `{.r-stack}` as well as the `{.fragment}`, for example with

```
:::: {.r-stack style="margin-left: 5%; margin-right: auto; width: fit-content;"}
::: {.fragment .fade-out fragment-index=2}
Content
:::
::: {.fragmment .current-visible fragment-index=2}
Something else
:::
::::
```

Slide Backgrounds (only in `revealjs`)

Set the `background` attribute on a slide to change the background color (all CSS color formats are supported).

Different background transitions are available via the `background-transition` option.



Learn more: Slide Backgrounds

Slide Backgrounds (only in `revealjs`)

When the background image is light, use the `.light-bg` tag to ensure that the slide text is in the normal `font-color`

- If the text is not very clearly visible, can always wrap the text in something like

```
::: {style="background-color: rgba(250, 250, 250, 0.7); border-radius: 0.8em 0.3em; padding: 0.3em;"}
Text
:::
```

(here, `rgba(250, 250, 250, 0.7` is a `#FAFAFA` background with opacity set to 0.7)



Media Backgrounds (only in `revealjs`)

You can also use the following as a slide background:

- An image: `background-image`
- A video: `background-video`
- An iframe: `background-iframe`



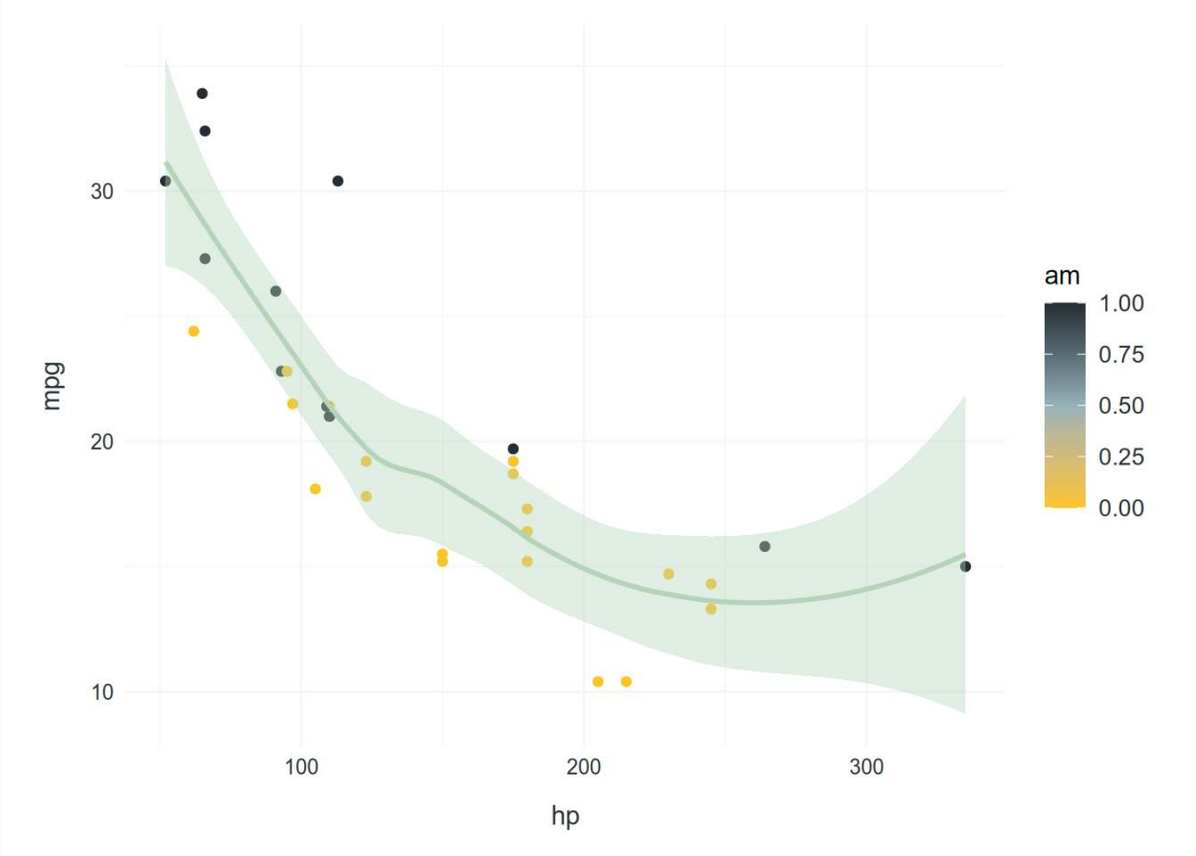
Slide Transitions

Possible options are the following:

Transition	Description
`none`	No transition (default, switch instantly)
`fade`	Cross fade
`slide`	Slide horizontally
`convex`	Slide at a convex angle
`concave`	Slide at a concave angle
`zoom`	Scale the incoming slide so it grows in from the center of the screen.

Tabsets

Plot | Data



Learn more: Tabsets

A long, scrollable table

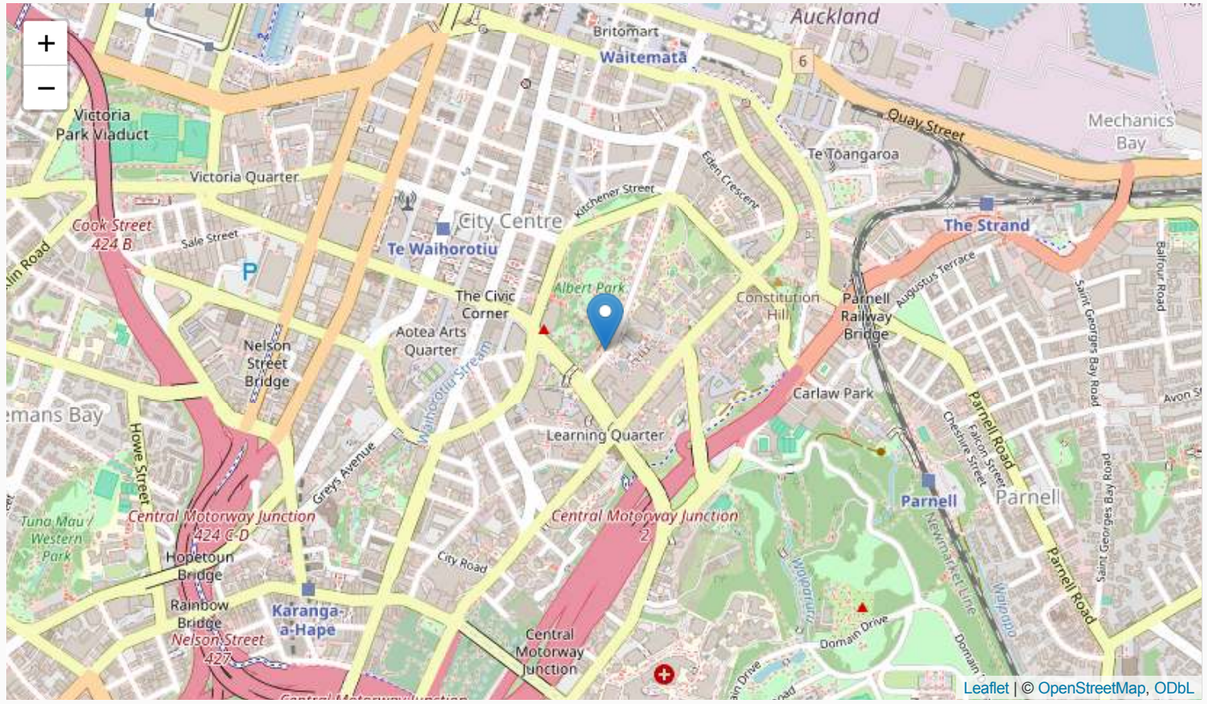
Using `styles`...

```
::: {style="height: 80%; overflow: auto; font-size: .65em;"}
Wrap the code to show the table here...
:::
```

mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4
21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4
22.8	4	108.0	93	3.85	2.320	18.61	1	1	4	1
21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1
18.7	8	360.0	175	3.15	3.440	17.02	0	0	3	2
18.1	6	225.0	105	2.76	3.460	20.22	1	0	3	1
14.3	8	360.0	245	3.21	3.570	15.84	0	0	3	4
24.4	4	146.7	62	3.69	3.190	20.00	1	0	4	2
22.8	4	140.8	95	3.92	3.150	22.90	1	0	4	2
19.2	6	167.6	123	3.92	3.440	18.30	1	0	4	4
17.8	6	167.6	123	3.92	3.440	18.90	1	0	4	4
16.4	8	275.8	180	3.07	4.070	17.40	0	0	3	3
17.3	8	275.8	180	3.07	3.730	17.60	0	0	3	3
15.2	8	275.8	180	3.07	3.780	18.00	0	0	3	3
10.4	8	472.0	205	2.93	5.250	17.98	0	0	3	4
10.4	8	460.0	215	3.00	5.424	17.82	0	0	3	4
14.7	8	440.0	230	3.23	5.345	17.42	0	0	3	4

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Thank you!


THANK YOU SO MUCH.